Python & AI Development Internship Final Report

Name: Rayan Badar **Company:** NexaSecure Tech

1. Weekly Summary

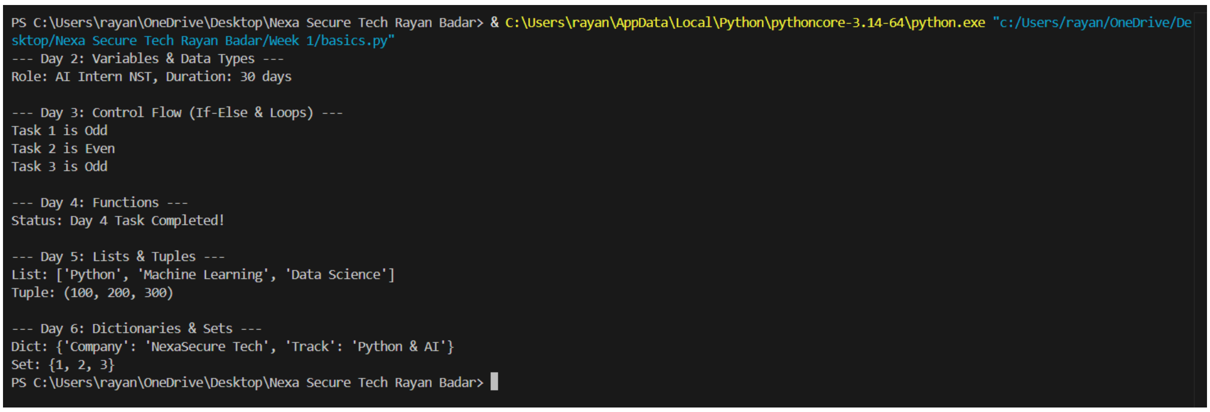
- **Week 1:** Set up Python and the IDE environment. Learned core programming concepts, including variables, loops, and data structures. Developed a Command Line Interface (CLI) Calculator.
- **Week 2:** Advanced into Python syntax, string manipulation, and file handling. Built interactive applications, including a To-Do List and a Temperature Converter. Introduced to Matplotlib for basic data visualization.
- **Week 3:** Explored Data Science concepts. Utilized Pandas for data manipulation and Matplotlib to visualize sample datasets with line, bar, and pie charts.
- **Week 4:** Built and trained Machine Learning models using Scikit-learn. Developed a Linear Regression model to predict student marks based on study hours, and trained an AI classifier using the Iris dataset.

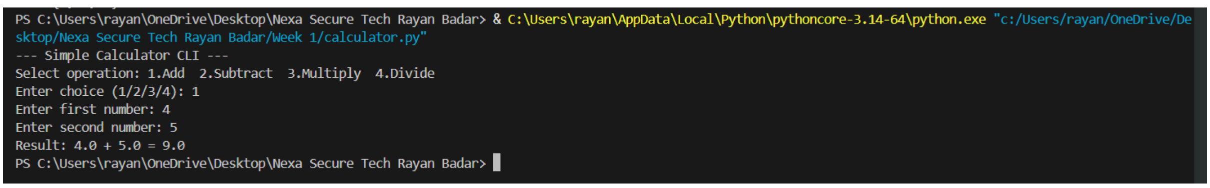
2. Skills Learned

- **Languages:** Python 3.x
- **Data Analysis:** Pandas, NumPy
- **Data Visualization:** Matplotlib
- **Machine Learning:** Scikit-learn (Linear Regression, Random Forest)
- **Tools:** VS Code, Git/GitHub version control

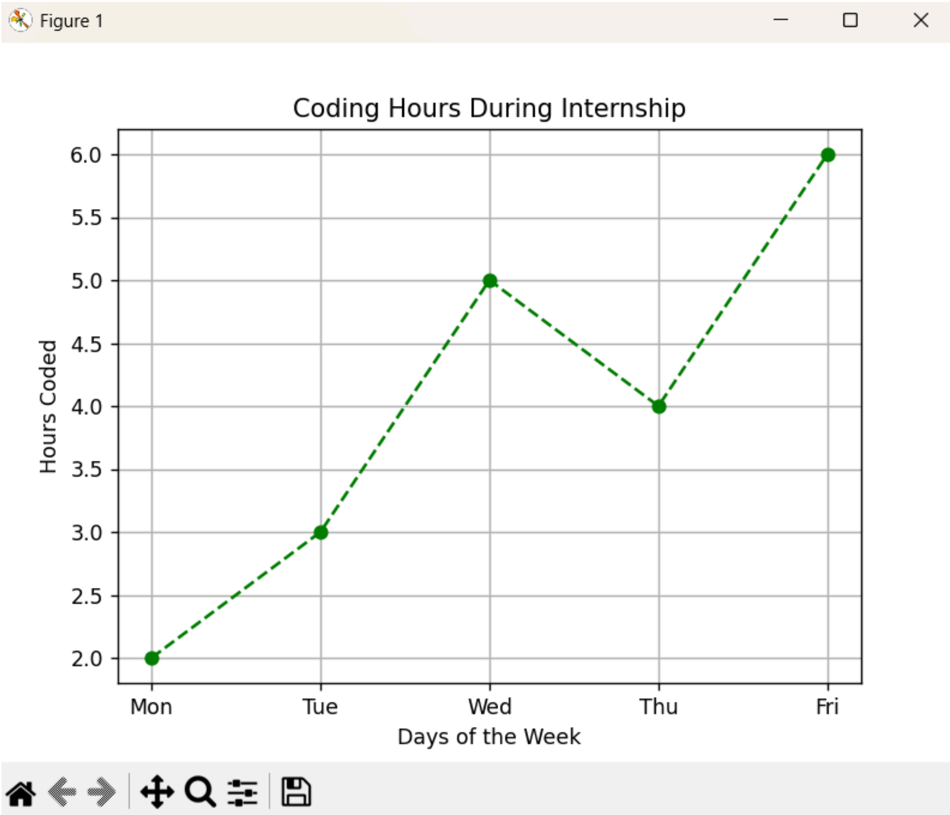
3. Project Outputs

Week 1 Outputs:





Week 2 Output:




```
PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar> & C:\Users\rayan\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/rayan/OneDrive/De
sktop/Nexa Secure Tech Rayan Badar/Week 2/temp_converter.py"
--- Mini Project 3: Temperature Converter ---

1. Celsius to Fahrenheit   2. Fahrenheit to Celsius   3. Exit
Choose conversion (1/2/3): 1
Enter temperature in Celsius: 34
 34.0°C is equal to 93.20°F

1. Celsius to Fahrenheit   2. Fahrenheit to Celsius   3. Exit
Choose conversion (1/2/3): 2
Enter temperature in Fahrenheit: 45
 45.0°F is equal to 7.22°C

1. Celsius to Fahrenheit   2. Fahrenheit to Celsius   3. Exit
Choose conversion (1/2/3): 3
Goodbye!
PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar>
```

```
PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar> & C:\Users\rayan\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/rayan/OneDrive/De
sktop/Nexa Secure Tech Rayan Badar/Week 2/todo.py"
--- Simple To-Do List App ---

Options:  1. Add Task   2. View Tasks   3. Exit
Enter your choice (1/2/3): 1
Enter the task description: Hello World
 'Hello World' added to your list.

Options:  1. Add Task   2. View Tasks   3. Exit
Enter your choice (1/2/3): 2

--- Your Tasks ---
1. Hello World
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Options:  1. Add Task   2. View Tasks   3. Exit
Enter your choice (1/2/3): 3
Exiting To-Do List App. Goodbye!
PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar>
```

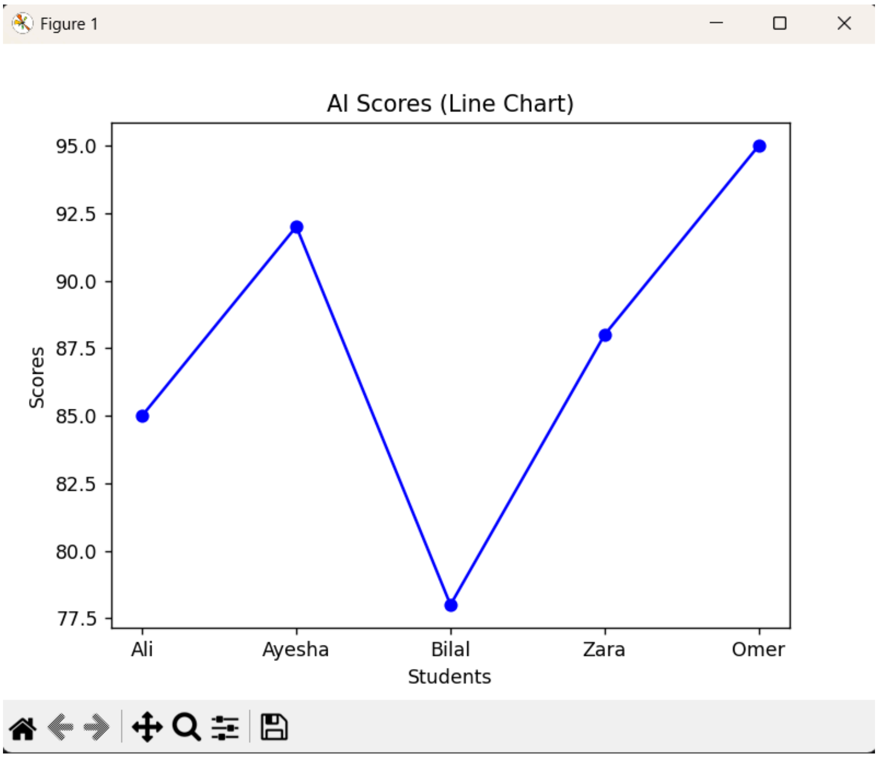
```
PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar> & C:\Users\rayan\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/rayan/OneDrive/Desktop/Nexa Secure Tech Rayan Badar/Week 2/week2_basics.py"
--- Day 8: String Operations ---
Original: '  NexaSecure Tech Internship  '
Cleaned & Upper: 'NEXASECURE TECH INTERNSHIP'

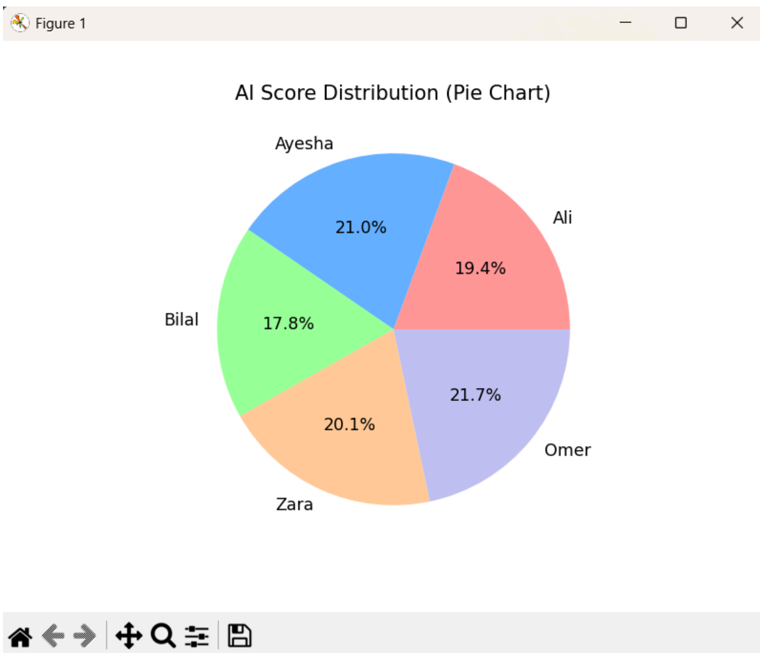
--- Day 9: File Handling ---
File Content: Learning Python & AI at NexaSecure!

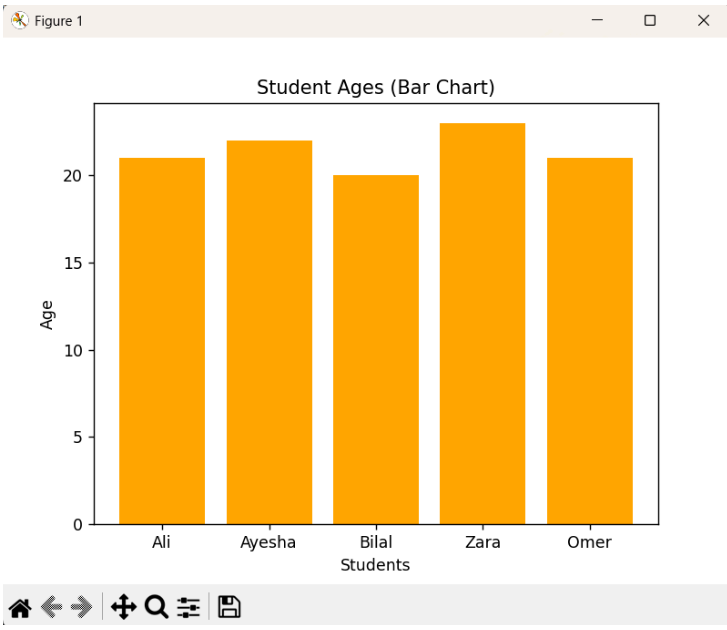
--- Day 10: Loops & List Comprehension ---
Original numbers: [1, 2, 3, 4, 5]
Squared numbers: [1, 4, 9, 16, 25]

PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar>
```

Week 3 Output:





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PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar> & C:\Users\rayan\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/rayan/OneDrive/Desktop/Nexa Secure Tech Rayan Badar/Week 3/week3_data_ai.py"
--- Day 16: NumPy Basics ---
NumPy Array: [10 20 30 40 50]
Array Mean: 30.0, Array Sum: 150

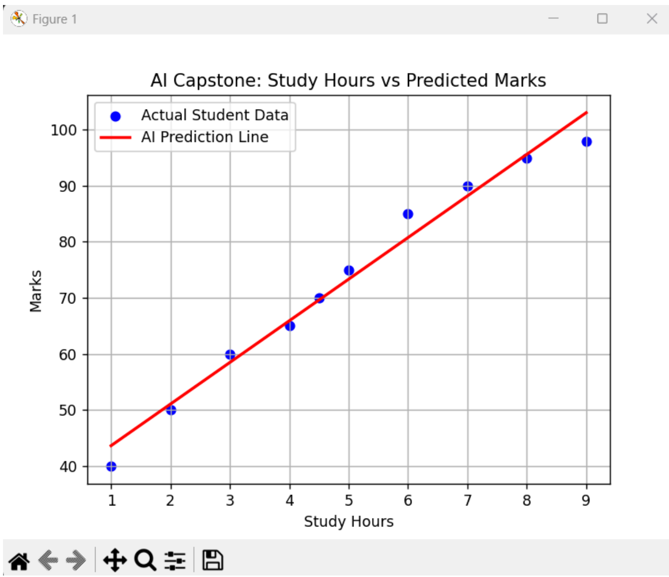
--- Day 17 & 18: Pandas & Mini Project 4 (Data Analysis) ---
 Dataset saved to 'students_data.csv'

Dataset Preview:
     Name  Age  AI_Score
0     Ali   21        85
1  Ayesha   22        92
2   Bilal   20        78
3    Zara   23        88
4    Omer   21        95

Average AI Score for the class: 87.6

--- Day 19 & 20: Matplotlib & Mini Project 5 (Visualizations) ---
 NOTE: Close each graph window to make the next one appear!
PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar>
```

Week 4 Output (Capstone):



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PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar> & C:\Users\rayan\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/rayan/OneDrive/Desktop/Nexa Secure Tech Rayan Badar/Week 4/capstone.py"
--- Day 23: Data Preparation ---
Dataset Head:
   Study_Hours  Marks
0          1.0     40
1          2.0     50
2          3.0     60
3          4.0     65
4          4.5     70

--- Day 24 & 25: Build and Test Model ---
 Model Trained Successfully!
C:\Users\rayan\AppData\Local\Python\pythoncore-3.14-64\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but
 LinearRegression was fitted with feature names
  warnings.warn(

 AI Prediction: If a student studies for 5.5 hours, their predicted marks are: 77.05

--- Day 26 & 27: Visualize Results ---
PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar>
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PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar> & C:\Users\rayan\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/rayan/OneDrive/Desktop/Nexa Secure Tech Rayan Badar/Week 4/iris_classifier.py"
--- Day 28: Mini Project 6 (Iris Flower Classifier) ---
Loading dataset...
Splitting data into training and testing sets...
Training the Random Forest AI Model...
Testing the model...

 AI Model Accuracy: 100.00%
The AI successfully learned how to classify the flowers!
PS C:\Users\rayan\OneDrive\Desktop\Nexa Secure Tech Rayan Badar>
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4. Conclusion

This 30-day internship at NexaSecure Tech successfully bridged the gap between foundational Python programming and practical Artificial Intelligence implementation. I transitioned from writing basic scripts to cleaning datasets, performing exploratory data analysis, and evaluating predictive machine learning models.